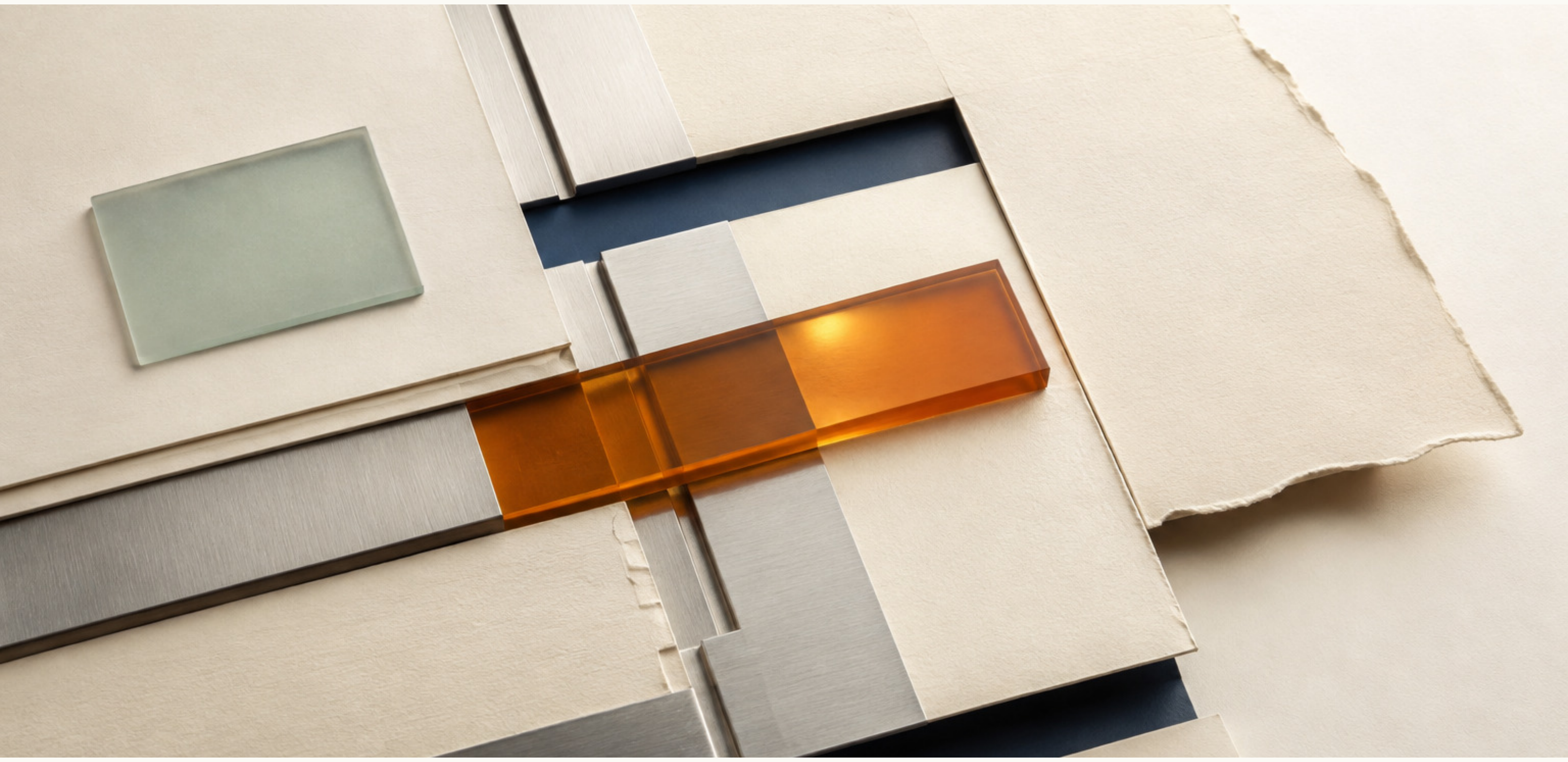
PAYMENTS / TREASURY / DIGITAL ASSETS

Accounting for an Omnichain Dollar

USDT0 reserve reconciliation: observed backing, accounting perimeter, and monitoring implications for banks.

PUBLIC-STATE CONCLUSION

Observed collateral and documented direct liabilities reconcile to 1.0003x. The measured difference is about three basis points; this report does not treat it as a reserve cushion.



TSOLMONDORJ NATSAGDORJ / SUWAPPU RESEARCH

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SUWAPPU.BOT/RESEARCH

BANKING TAKEAWAY

Observed backing reconciles to par. That is not the same as a reserve attestation.

At 01:53 UTC on 1 August 2026, the verified Ethereum lockbox held $3.4536B of USDT against $3.4526B of directly measured USDT0 liabilities across the issuer-documented perimeter. Observed coverage was 1.0003x. The $1.03M arithmetic difference - about three basis points - is smaller than risks the balance read cannot observe, so we classify the result as reconciled to par rather than excess reserve coverage.

$3.4536B
OBSERVED RESERVE BALANCE

1.0003x
OBSERVED COVERAGE RATIO

~3bp
MEASURED DIFFERENCE / NOT A CUSHION

WHAT A BANK CAN AUTOMATE

The public-state control can read the canonical reserve account, aggregate every documented direct liability leg, version the accounting perimeter and alert on mismatch. The 12-month panel provides 183 block-aligned observations; the current head check covers all 20 documented direct liability legs in one session.

WHAT STILL REQUIRES EXTERNAL EVIDENCE

A token balance does not establish whether reserves are encumbered, whether net mint/burn messages are in flight, or whether the deployment registry itself is complete. A bank also needs governance around the address map and independent evidence for the legal and operational status of reserves. At a 3bp margin, these are decision-relevant, not footnotes.

CONTROL CONCLUSION

Use public-state reconciliation as a continuously repeatable first-line control. Treat registry governance, message state and reserve encumbrance as separate evidence requirements before relying on the asset for treasury, settlement or client-facing flows.

CONTROL FINDING

The two corrections are the control finding.

The first error created false excess coverage; the second created a false reserve shortfall. Both came from reference-data and perimeter failures that a bank control framework should catch before a ratio reaches a risk committee.

VERSION	REPORTED SIGNAL	INTERPRETATION	CONTROL FAILURE
V1	**1.042 endpoint**	**False surplus**	Liability perimeter truncated
V2	**0.513-0.588**	**False shortfall**	Wrong Polygon collateral address
V3	**1.0003 head**	**Par / ~3bp**	Complete documented universe

CONTROL IMPLICATIONS

01
Liability-perimeter governance

V1 omitted $134.8M of liabilities. Completing the universe removed 96% of the reported buffer. Scope needs an owner, effective date, inclusion rule and completeness check.

02
Canonical-account governance

V2's control address read $0.02 while the canonical Polygon predicate held $1.22B-$1.39B. In a monitoring system, that is a reference-data failure, not reserve deterioration.

03
Timestamp discipline

Cross-chain balances need a common observation time. The panel resolves each chain to the target timestamp; the migration window is rescanned every six hours so settlement flow is not misread as reserve deficiency.

OPERATING RULE

Treat the address registry as controlled reference data: version scope, canonical contracts, effective dates and containment relationships before calculating coverage.

OBSERVED PERIMETER

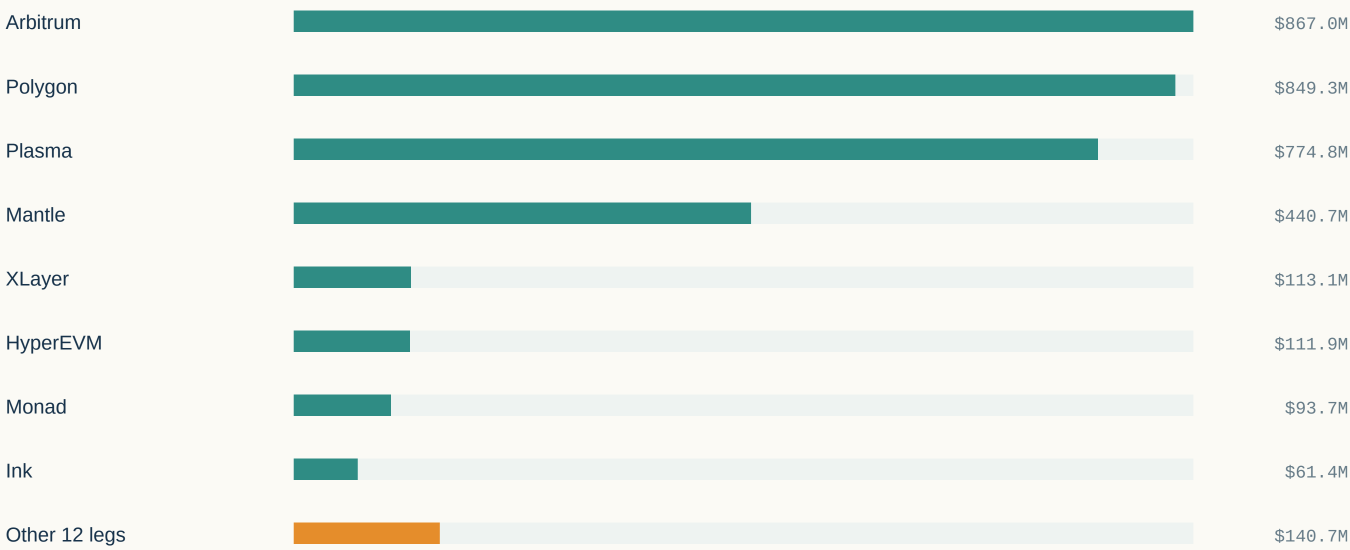
The measurable perimeter closes across twenty direct liability legs.

The observable accounting identity is straightforward: the verified Ethereum reserve balance should cover aggregate supply across the issuer-documented direct USDT0 deployments. HyperCore is verified as a sub-ledger of HyperEVM and is not double-counted; Tron and TON sit outside this perimeter as Legacy Mesh.

1 AUG 2026 / 01:53 UTC

$3.4536B collateral / $3.4526B liabilities = 1.0003x

DIRECT LIABILITY PERIMETER



20 legs directly measured **0 excluded** **1 verified sub-ledger** HEAD_SNAPSHOT_20260801.JSON

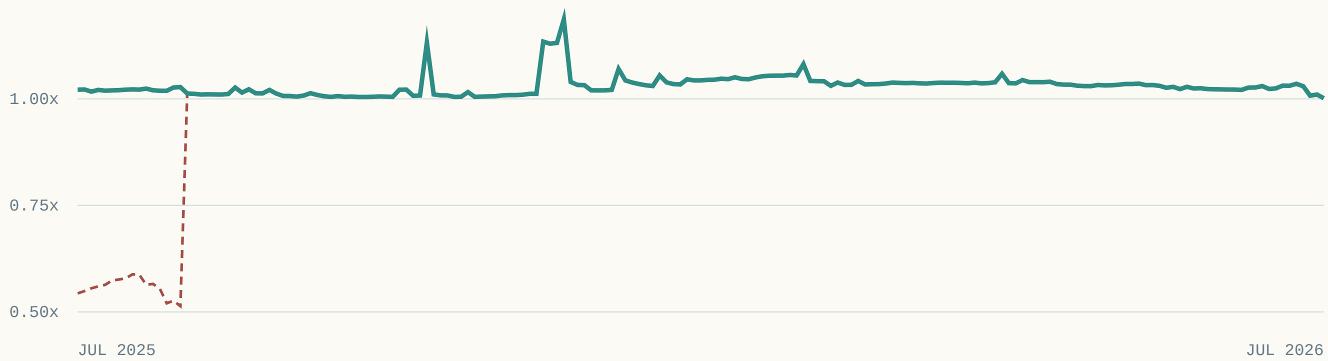
INTERPRETATION

For a bank control, 1.0003x should be classified as reconciled to par within measurement tolerance, not as positive excess reserves. The head reads span roughly a minute rather than one aligned block and do not constitute a reserve attestation.

WHY THE SHORTFALL DISAPPEARED

$1.3B of apparent reserve deficiency was address-mapping risk.

Before 27 August 2025, Polygon PoS USDT backing sat in the canonical Polygon predicate rather than the USDT0 lockbox. V2 checked a different address and converted an address-map failure into an apparent 41%-49% reserve deficiency. For a bank monitor, wrong reference data can be as consequential as a wrong balance.



DASHED: V2 PUBLISHED   SOLID: CANONICAL PREDICATE INCLUDED

0 / 183 CORRECTED OBSERVATIONS BELOW PAR

SIX-HOUR MIGRATION BRACKET / 27 AUG 2025

-$1,358.8M	+$1,258.6M	-$98.1M	$2.1M
CANONICAL PREDICATE	ETHEREUM LOCKBOX	ARBITRUM SUPPLY	RESIDUAL FLOW

CONTROL VALIDATION

The $1,358.8M predicate outflow matched Polygon supply at bracket open ($1,358.759M) to within about $82K - 0.006% of the flow. The issuer also announced the backing migration that day. Independent documentary evidence and chain state describe the same perimeter change.

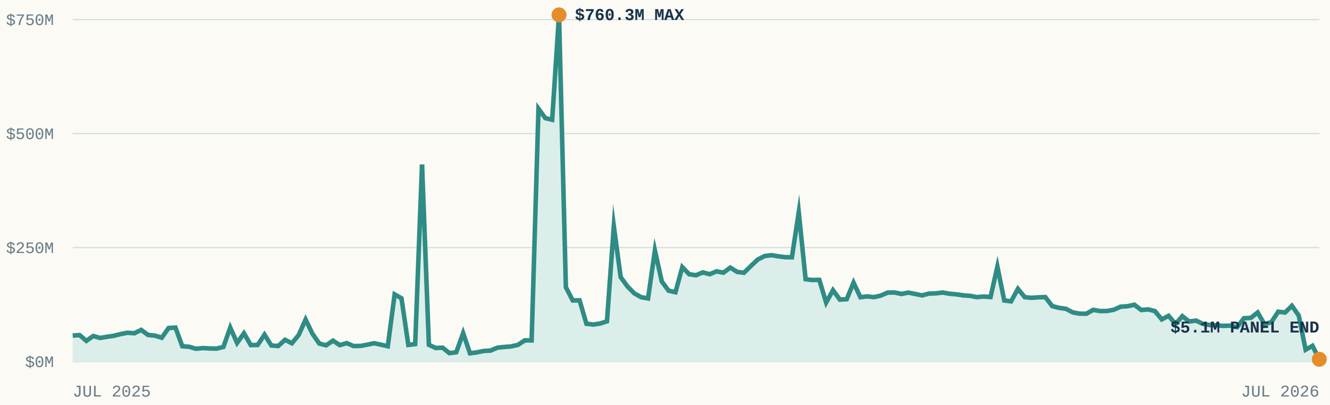
LESSON

Treat contract and address changes as governed reference-data events. Otherwise an accounting-boundary migration can look indistinguishable from a reserve impairment.

HISTORICAL CONTEXT

Coverage moved from a visible cushion to measurement tolerance.

Historical corrected coverage stayed above par, but the measured excess was not stable: the buffer share ranged from roughly 15bp to 18.7% of liabilities. By the end of the sample, excess coverage had compressed to measurement tolerance. For a bank, the endpoint matters more than the historical maximum.



CORRECTED COLLATERAL MINUS DOCUMENTED LIABILITIES

$760.3M
MAX / 15 DEC 2025

$296.9M
31 DEC 2025

$122.4M
17 JUL 2026

$5.1M
25 JUL / PANEL END

FINAL EIGHT DAYS

-$192.9M collateral vs -$75.7M liabilities

About $117M more collateral left than liabilities declined. The panel buffer fell to $5.1M; the complete-universe head check six days later measured $1.03M.

INTERPRETATION

For underwriting, historical excess coverage should not be carried forward as a current risk buffer. The relevant state is the endpoint: observed coverage is effectively par, and unobservable exposures are larger than the measured difference.

PUBLIC-STATE CHECK VS ATTESTATION

The ledger can be reconciled; the reserve claim still needs off-chain evidence.

Public chain state can support a repeatable first-line reserve control. It cannot, by itself, answer the legal and operational questions a bank needs before treating that control as proof of available reserves.

AUTOMATABLE ONCHAIN CONTROL

- **Collateral balance**
  USDT balanceOf() on the verified Ethereum OAdapter lockbox.
- **Documented token supply**
  totalSupply() on each documented USDT0 deployment, with decimals and live contracts checked.
- **Versioned accounting perimeter**
  Twenty direct liability legs, plus an explicit HyperCore containment check; Tron and TON classified as Legacy Mesh.

REQUIRES ADDITIONAL EVIDENCE

- **Asset encumbrance / legal claim**
  A positive token balance does not establish that reserves are unencumbered, bankruptcy-remote, or senior to other claims.
- **Settlement state / messages in flight**
  A point-in-time sum does not expose net mint or burn instructions moving between chains.
- **Registry completeness / governance**
  The method verifies documented deployments; it cannot prove that the issuer registry omits nothing or that its change process is complete.

MINIMUM EVIDENCE STACK FOR A BANK

01 Versioned liability registry
02 Canonical reserve accounts
03 Net messages in flight
04 Encumbrance / legal-status evidence
05 Contract + migration change control
06 Independent attestation / registry check

THRESHOLD

At ~3bp, any omitted, in-flight or encumbered position above $1.03M overwhelms the arithmetic difference. Escalation policy should reflect that scale.

SUGGESTED OPERATING MODEL

Automate the reconciliation; govern the evidence it cannot see.

The practical architecture is a layered control: public-state reconciliation for continuous observation, event-driven governance for perimeter changes, and external evidence for reserve conditions that token balances cannot establish.

CONTROL LAYERS

01

Daily public-state reconciliation

Read canonical reserves and every documented direct liability leg; version the timestamp, address map and accounting perimeter; alert when a configured threshold is breached.

02

Event-driven perimeter controls

Detect new deployments, migrations, upgrades, containment changes and registry revisions. Reverify contract identity, symbol, decimals and supply before inclusion.

03

External evidence overlay

Pair the chain-state control with issuer or attestation evidence for encumbrance, registry completeness and message state; escalate when unobservable exposure can exceed the measured difference.

INDEPENDENT AUDIT TRAIL

data/usdt0_timeseries.csv	183-row panel + retained wrong-address control
data/polygon_predicate_prebreak.json	canonical Polygon backing backfill
data/head_snapshot_20260801.json	complete-universe head measurement
data/buffer_dynamics.json	flow coupling + terminal drawdown
code/collect_usdt0.py	panel collection harness
code/head_snapshot.py	current-universe snapshot

OPEN METHODOLOGY AND SOURCE DATA

suwappu.bot/research/replication

CONCLUSION

The underwriting upgrade is a controlled accounting perimeter.

The durable result is not the 1.0003x ratio. It is that a cross-chain dollar can be monitored as a defined set of reserve and liability accounts - provided the perimeter itself is governed like financial reference data and the off-chain evidence remains separate.

01

For treasury and payments

Use the public-state check as a repeatable control overlay, not an attestation. It can flag divergence or stale reference data before the asset is relied on for settlement or client-facing flows.

02

For risk and model governance

Version the registry, address map and method. Treat contract migrations, new deployments and corrections as controlled model-data changes with an auditable history.

03

For issuers and infrastructure partners

Publish a dated machine-readable registry of each liability leg, backing account, containment relationship and migration. This reduces operational and model risk for counterparties.

DISCLOSURES

This report is research, not a reserve attestation, audit opinion or investment recommendation. Suwappu builds cross-chain execution infrastructure and holds operational stablecoin balances, including USDT and USDT0, incidental to running it. No directional position informed this analysis. Tether, Everdawn Labs, and other named parties did not review the work before publication. All inputs are public chain state or cited public documents. The working paper is canonical where this report and the paper differ.

SELECTED SOURCES

USDT0 technical documentation: deployments and Legacy Mesh (accessed 31 Jul 2026)

USDT0, Polygon USDT Now Upgraded to USDT0 (27 Aug 2025)

Polygon PoS Bridge documentation: canonical ERC20 predicate (accessed 31 Jul 2026)

Everdawn Labs: USDT0 audit reports; Chaos Labs: USDT0 Mechanism Design Review (2025)

Newey-West (1987); Politis-Romano (1994); Bai-Perron (1998)

WORKING PAPER + EVIDENCE BUNDLE

SUWAPPU.BOT/RESEARCH/REPLICATION